

PERSONAL INFO

Birthdate: 01.09.1986

Citizenship: Germany

CONTACT

+49 176 22598542

mihail.balanici@gmail.com

Berlin, Germany

LinkedIn

Google Scholar

SKILLS

- Data Engineering / DevOps
- Machine Learning / MLOps
- ETL and ML Pipelines
- Network Engineering & Automation
- CI/CD
- Data Center Networks
- Optical Networks and Systems
- Research & Development
- Scientific Writing
- Project Management

PROGRAMMING

- Python
- Matlab
- SQL, NoSQL, Time Series Databases
- Shell scripting
- PLC programming

MIHAIL BALANICI

ELECTRICAL, DATA & ML ENGINEER / SCIENTIST

Electrical Engineer & Research Associate at Fraunhofer Heinrich Hertz Institute (HHI) in Berlin, specialising in Communications and Network Engineering, Data Engineering, Machine Learning and Data Science.

Currently working on R&D of Network Automation solutions by developing scalable Telemetry frameworks and ML pipelines, using the latest state-of-the-art technologies and ETL tools. Experienced in the fields of Data Center Networking/Cloud Computing, which represent the topic of my PhD research.

WORK EXPERIENCE

(More details on LinkedIn)

Research Associate / Scientist

2021 - PRESENT
Berlin, Germany

Fraunhofer Heinrich Hertz Institute (HHI)
Department of Photonic Networks and Systems

- Research and development of telemetry frameworks, data analytics, and ETL & ML pipelines for network and service automation, traffic monitoring and analytics, and autonomous resource allocation in optical networks
- Developed from scratch, deployed and made key contributions to the first prototype Technology Demonstrators and Proof-of-Concepts (PoC) for industry and national research projects, as well as standardization proposals
- Presented and published the developed prototypes & PoCs at international conferences, journals, workshops, exhibitions, and project meetings

Doctoral Research Assistant

2016 - 2020
Kiel, Germany

Kiel University (CAU)
Institute of Electrical Engineering and Information Technology

- Research of Hybrid Electro-Optical Intra-Data Center Networks (DCN) and their optimization using traffic modeling, machine learning for traffic forecasting, and optical circuit switching (OCS)
- Prepared and taught exercise and lab courses on Digital Communications, Fiber-Optic Communication Networks, and Communications Lab

System Engineer

2014 - 2015
Ulm, Germany

ES Engineering and Service GmbH

- Software development, integration and testing for industrial control systems and industry automation (Siemens SIMATIC S7 PLC, IoT sensors', and HMI displays' programming)
- Design and preparation of electrical documentation (ESplan CAD/CAE)
- Commissioning and technical deployment at the client's location

Graduate Student Researcher

2013 - 2014
Ulm, Germany

Ulm University
Institute of Distributed Systems

- Master thesis research on development of a cluster-based data aggregation protocol (CBAP) for V2V Communication / Vehicular Ad-hoc Networks
- Published key results at Vehicular Networking Conference (VNC) 2013, Boston, MA, USA

TOOLS

- Linux, Windows, MacOS
- Telegraf, InfluxDB, Chronograf, Kapacitor (TICK stack)
- MongoDB, Redis
- Grafana
- Apache Kafka, Apache Spark
- Docker, Kubernetes
- Protocols: NETCONF/YANG, SNMP/MIB, gRPC/gNMI
- Data formats: JSON, XML, Protocol Buffers
- ML Packages: PyTorch, TensorFlow, scikit-learn, etc.
- Standard Commands for Programmable Instruments (SCPI)
- Git
- MS Office, MS Visio
- Latex, OriginPro

LANGUAGES

- English (Near native proficiency)
- German (Near native proficiency)
- Russian (Near native proficiency)
- Romanian (Native language)

LICENSES

- Driver's License B Class
- Sport Boating License (SBF-See & SBF-Binnen)

INTERESTS

- Fitness & Tennis
- Investing & Financial Markets
- Digital Photography
- Traveling and Hiking

EDUCATION

Dr.-Ing., Electrical Engineering and Information Technology — **2016 - PRESENT**
Kiel University (CAU) — Kiel, Germany
Doctoral Thesis: *Machine Learning-based Traffic Forecasting for Hybrid Electro-Optical Intra-Data Center Networks* (currently in finalization)

M.Sc., Communications Engineering — **2011 - 2013**
Ulm University — Ulm, Germany
Master Thesis: *Cluster-based Aggregation for Inter-vehicle Communication*

Master, Electrical Systems and Communications — **2009 - 2011**
Technical University of Moldova — Chisinau, Moldova
Master Thesis: *Migration of Data Communication Networks based on Traditional Technologies to the IP/MPLS Data Transfer Technology*

Cisco Certified Networking Associate (CCNA) Trainee — **2010 - 2011**
Cisco Networking Academy — Chisinau, Moldova

Bachelor, Electronics and Communications — **2005 - 2009**
Technical University of Moldova — Chisinau, Moldova
Bachelor Thesis: *Radiative Recombination in Layered MoS2 crystals*
Graduated with honors (1st among 85 graduates)

KEY PUBLICATIONS

(Full list on Google Scholar)

- M. Balanici et. al., "*Autonomous Link-Capacity Adjustment using TeraFlowSDN Controller in a Disaggregated Optical Network Testbed*," in European Conference on Optical Communication (ECOC), D4 Demo, Frankfurt am Main, Germany, September 2024.
- M. Balanici et. al., "*Live Demonstration of Autonomous Link-Capacity Adjustment in Optical Metro-Aggregation Networks*," in Optical Fiber Communication (OFC) Conference, M3Z.3, pp. 1-3, San Diego, CA, USA, March 2024.
- M. Balanici et. al., "*Demonstration of a Real-Time ML Pipeline for Traffic Forecasting in AI-Assisted F5G Optical Access Networks*," in European Conference on Optical Communication (ECOC), Tu2.5, pp. 1-4, Basel, Switzerland, September 2022.
- M. Balanici et. al., "*Classification and Forecasting of Real-Time Server Traffic Flows Employing Long Short-Term Memory for Hybrid E/O Data Center Networks*," in IEEE/OSA Journal of Optical Communications and Networking (JOCN), vol. 13, no. 5, pp. 85-93, May 2021.
- M. Balanici et. al., "*Machine Learning-based Traffic Prediction for Optical Switching Resource Allocation in Hybrid Intra-Data Center Networks*," in Optical Fiber Communication Conference (OFC), Th1H.4, pp. 1-3, San Diego, CA, USA, March 2019.
- M. Balanici et. al., "*Hybrid Electro-Optical Intra-Data Center Networks Tailored for Different Traffic Classes*," in IEEE/OSA Journal of Optical Communications and Networking (JOCN), vol. 10, no. 11, pp. 889-901, November 2018.

AWARDS

DAAD Scholarship | German Academic Exchange Service (DAAD) | 2011 - 2013

Orange Scholarship | Orange Moldova | 2009

Presidential Medal for Excellency | President of the Republic of Moldova | 2009

Moldcell Scholarship | Moldcell, TeliaSonera Group | 2008